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Research Topic: REVIVER: Smart Filtration System Installed at Major Sources of Microplastic Pollution to Reduce Biological Impacts in Marine Ecosystems

Title: REVIVER: Smart Filtration System Installed at Major Sources of Microplastic Pollution to Reduce Biological Impacts in Marine Ecosystems

Research Problem

Marine environments worldwide are threatened by microplastic pollution. The particles can enter the environment either directly or in the form of fragmented pieces of larger plastic objects. Waste water and other land-based pollutants transport microplastic into rivers and coastal areas. However, existing waste water treatment methods do not remove all the particles from the waste water. There are reports of physiological effects in different marine species and organisms due to microplastic ingestion, but there is little knowledge of the environmental exposure, the trophic transfer of microplastics and the potential risks for higher-trophic-level species. [1]

Whales are key organisms to study the wider biological impacts of marine pollution. Because of their long lifespan, wide ranging movements and high position in food webs, they are highly susceptible to environment persisting pollutants. Therefore, whales are relevant to study to determine the greater effects of microplastics on organisms of many trophic levels in the food webs.

In addition to their environmental value, whales have recently become a symbol of intelligence, social behavior, communication, and of the connection to the ocean which humans lack. They have appeared in multiple films, documentaries, and other media. A prominent television drama, for example, featured whales as recurring visual motifs to get more attention to their ecological niche. [2] Given their significance in nature and their popularity within the public, the inclusion of cetaceans as a higher-trophic-level species allows for an assessment of the broader ecological benefits that could be obtained from reducing microplastics in the oceans.

This proposal aims to investigate whether intercepting microplastics at sources can prevent their dispersal, subsequently reducing exposure to marine life, and potentially reduce biological risks towards marine organisms (specifically cetaceans).

‘REVIVER’ is proposed as a smart, source-based filtration system that combines multi-stage filtration, particle detection, AI-assisted classification, and environmental monitoring.

Literature Review:

Previous research has established microplastics as persistent contaminants within marine ecosystems, with a broad range of organisms from different trophic levels having been shown to ingest them. In addition to potential biological effects which have been reported to include oxidative stress, altered metabolism, inflammation, altered feeding and reproductive problems, it can also travel to predators through the food web. [3][4]Removing microplastic particles from wastewater by means of membrane filtration, coagulation/sedimentation and adsorption have been studied, but has not yet led to the complete removal of all microplastic particles, as the methods for detection of these small particles are not yet standardized. [5][6]Whilst there is an abundance of work on filtration and the marine biological effects of microplastics, and a number of cetacean monitoring programs in place around the world [7][8], there is currently no source-based system able to intercept microplastics and inform of potential ecological benefits. ‘REVIVER’ is proposed to address this gap through an integrated technological and ecological assessment.

There are three themes identified in the scientific studies done on the transport of microplastics to aquatic environments via wastewater. In the first place is the major theme of this topic, which covers the process that microplastics take part in, entering the ecosystem through the wastewater. Liu et al. conducted a review of the global techniques used to remove microplastics from wastewater. The mechanisms of removing microplastics from wastewater were discussed with examples from recent studies and also included filter technology used in the removal of microplastics along with a discussion of the difference between the mechanism of removal and the behavior of plastic materials in the process of removal [1]. The authors have also discussed the different approaches of treatment including filtration, adsorption, and coagulation and the difficulties related to their effectiveness measurement. [7]

The chosen organism, cetaceans, are very likely to be exposed to microplastics both in the seawater and in their prey. In a systematic review of the research on the presence of microplastics in marine life forms, it is established that there is no consistency in terms of sampling method and data presentation in different organisms, therefore direct comparisons cannot be made [6]. In the case of the baleen whales like blue, fin and humpback whales, there is a lot of exposure through filter feeding and high prey consumption.

Saha and Shourie highlighted out the analysis of complex wastewater matrices by current sampling methods and analysis techniques faces several challenges, such as a lack of standard methods for analysis [9]. A study by Mathew et al. identified the need for proper pre-treatment and quality control to detect the microplastic particles in wastewater samples [10]. That will be addressed in this study using controlled samples, contamination control measures, repeating experiments and evaluating the performance of particle detection.

According to Zantis et al., the exposure to microplastics in baleen whales was calculated based on the information about the fecal matter and feeding behavior of these animals. As shown by the study, the average daily exposure rate in baleen whales is equal to 3.4 million microplastic particles depending on diet and geographic location [11].

In another study, Kahane-Rapport et al. used data on microplastics in combination with foraging patterns of whales and established that 99% of microplastics intake by blue, fin, and humpback whales takes place through trophic transfer, with daily intake of 10 million and 200,000 microplastics in krill-eating blue and humpback whales, respectively [12].

Daily intake of tens of thousands of synthetic particles in North Atlantic fin whales associated with the consumption of krill was reported by Garcia-Garin et al. [13].

In toothed whales, Moore et al. found microplastics in all 7 beluga whales from the eastern Beaufort Sea and described the likely route of exposure but had no information on health effects. In order to better understand the relevant biology of whales in relation to filtration, more work is required [14].

In relation to REVIVER, there is a logical and applicable link to the studies of whales' biology, namely the reduction of microplastics from wastewater and other water bodies will alleviate loading of microplastics in marine food webs. However, the aim of this project is not to prove that filtration would bring about health benefits to whales. The significance of whales will be evaluated as a reason to prioritize source reduction of microplastics [6], [12]-[17].

Research Question

How would decreasing the microplastic input by a contained, modular, filtration-and-monitoring system impact the biological health of marine organisms and higher-trophic-level species like cetaceans?

Sub-questions:

1. How does the efficiency of REVIVER vary with the size of the particles, the flow rate of water, and the configuration of the filters?

2. To what extent can the monitoring aspect of REVIVER differentiate between microplastics and non-plastics?

3. How could the REVIVER's decrease in microplastics at the source impact the exposure of microplastics to marine food webs and cetaceans?

4. What would be the potential ecological and global impact of REVIVER on a broad scale, and what are the limiting factors that prevent the large-scale use of REVIVER?

Methodology

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This study will involve a mixed-methods design, in order to both assess the quantifiable performance of REVIVER, plus measure the potential ecological importance of reducing microplastic inputs to the marine environment. It will involve laboratory-scale experimental prototypes, and a literature review of marine organisms and cetaceans, with live animals not being subject to experiments.

Quantitative component:

Samples of filtered sea water will be taken and tested where there has been a predetermined concentration of microplastics introduced. One arrangement of a lab-scale REVIVER prototype will be developed consisting of multiple stages of filtration including pre-screening, fine/membrane filtration, particle detection chamber and closed collection unit. Controlled samples of water with known amount of microplastic particles of known size will be undertaken through various arrangements of filters. The quantity of microplastics in the samples will be determined before and after filtration enabling the percentage of particles removed to be determined for various configuration. The size and number of particles, flow rate, filter pressure, turbidity, and other relevant environmental conditions will be measured and recorded for each trial. The system for detecting particles will be tested using optical imaging and computer-vision techniques to ascertain the extent to which particles of microplastics can be distinguished from organic material and from sediment. Experimental results will be analyzed to determine implications for practical implementation including energy consumption, pressure development, fouling, and maintenance.

Qualitative component:

A literature review will be conducted to identify the biological effects of microplastics in cetaceans in terms of their influence on nutrition, feeding, growth, reproduction, stress response, and cell integrity. The literature search will be focused on whales, dolphins, and porpoises to establish how these animals are exposed to microplastics environmentally and through their diet. A marine food web will be built in order to determine how any change in microplastics concentration in one trophic level may affect the flow of microplastics in other trophic levels. The resulting findings will be compared to establish the biological and ecological impacts that can be prevented through a decrease in microplastic inputs in the environment. The qualitative findings will be combined with prototype findings in order to determine if the decrease in microplastic inputs is likely to have any significance for marine life and its predators.

Ethical considerations:

No cetaceans or any other form of live marine life will be captured, experimented on or exposed to microplastics during the study. Microplastic samples used in laboratory experimentation will be retained as part of the experiment and disposed of or returned to circulation post experiment. Care will be taken to avoid secondary contamination during the process of sampling and experimentation. Where any form of biological data is used in the study, it will be appropriately cited as it will be based on existing research. Any further experimentation using biological samples would only be done with appropriate permission from the relevant institutions.

Expected outcomes and limitations

The expected primary product of this work is a quantitative analysis comparing filter configurations and the resulting particle removal in relation to size and flow factors. A secondary outcome is an appraisal of the degree and limitations of the monitoring component's classification ability. A third result is a literature-based ecological implication of the source reduction on potential downstream benefit in exposure. The latter is of particular relevance to baleen whales, since recent studies have demonstrated that exposure for these species may be largely determined by the microplastic concentration in their prey, rather than by direct filtration from ambient water [12], [13]. The project will clarify the distinction between: (1) what REVIVER shows in a laboratory setting, (2) what published research shows regarding whale exposure, and (3) what may be inferred regarding the benefits of source reduction.

The study has significant limitations. First, a laboratory water loop is unable to replicate the chemical and biological complexity of wastewater and optical classification cannot substitute polymer confirming methods such as spectroscopy. A small prototype is unable to demonstrate city or country-level removal. Second, estimates of whale exposure are highly dependent on the species, diet, habitat, particle concentration and modeling assumptions [12], [13]. Evidence of the ingestion of microplastics by whales and other cetaceans alone do not prove that microplastics leads to population declines or specific disease; the literature continues to identify

significant uncertainties in the process of exposure and health effect [6], [15]. The project will therefore use evidence from whales to demonstrate ecological relevance, rather than a demonstrated causal health outcome.

Conclusion

REVIVER offers a focused, testable approach for microplastic source-based microplastic reduction. It focuses on the removal of individual particles at the source and thus an evaluation of the performance and trade-offs of the technology. The literature will be reviewed and synthesized in order to predict its potential ecological consequences. Last, by distinguishing between variables demonstrated and those deduced from the literature, the scientific and ethical integrity of the work is sustained while pursuing an innovative endeavor that simultaneously incorporates source reduction and monitoring.

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